

Who Lives Where: Linking Biodiversity Observations to Cadastral and Taxonomic Knowledge

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Abstract. This paper presents Who Lives Where, a linked data pipeline for the Dutch National Databank Flora and Fauna (NDFF), applied here to the Twente region. We triplify NDFF observation features using Darwin Core, Dublin Core Metadata Terms, and GeoSPARQL, then extend the dataset with a custom OWL 2 ontology for fields unrepresented in existing vocabularies: conservation policy status, monitoring protocol, grid cell identifiers, location blurring levels, and species group. The ontology layer is linked to the occurrence layer through shared scientific name values. A federated query architecture links the graph to Wikidata for taxonomic hierarchy traversal and to the Kadaster Knowledge Graph for Dutch administrative geometries via SPARQL `SERVICE` and GeoSPARQL. The resulting graph enables conservation-aware spatial queries that combine NDFF occurrence data with taxonomic hierarchy and cadastral geometry. The source code is available on GitHub¹.

Keywords: Darwin Core · OWL ontology · Linked Data · GeoSPARQL · Wikidata

1 Introduction

The Dutch National Databank Flora and Fauna (NDFF)² holds over 50 million species occurrence records at full coordinate precision, yet it is not published as linked data and cannot be queried alongside infrastructure such as the Kadaster Knowledge Graph (KKG)³, which contains Dutch base registers for addresses, buildings, and topography in RDF form. Many species are legally protected under the *Wet natuurbescherming*, and their presence at a development site carries legal force. Bridging this gap would enable queries such as identifying the nearest recorded observation of a protected species to a given building address, determining which municipalities show the highest species diversity, or assessing whether the distribution of Habitats Directive species correlates with recent construction activity.

¹ <https://github.com/2526-LDSW-G9>

² <https://ndff.nl/>

³ <https://data.kkg.kadaster.nl/>

This paper investigates how biodiversity occurrence data can be triplified, ontologically enriched, and federated with cadastral and taxonomic knowledge graphs to support spatial and policy-driven queries. We present *Who Lives Where*, a linked data pipeline for NDFF observation data covering the Twente region of the Netherlands. The contributions are: (1) a triplification pipeline using Darwin Core, Dublin Core Metadata Terms, and GeoSPARQL, producing a full-resolution occurrence triple dataset; (2) a domain ontology covering conservation policy status, monitoring protocol, location blurring, and species group, all fields unrepresented in existing biodiversity vocabularies, validated with the HermiT OWL reasoner; (3) a federated query architecture linking the graph to Wikidata for taxonomic hierarchy traversal and to KKG for Dutch cadastral geometries via SPARQL SERVICE and GeoSPARQL; and (4) SHACL validation of the integrated graph.

Section 2 reviews related work on biodiversity linked data and ontology engineering. Section 3 describes the full pipeline, from dataset selection and triplification through ontology design, federated integration, and validation. Section 4 presents the results of the analytical queries. Section 5 discusses the findings and outlines directions for future work.

2 Related Work

Prior work spans biodiversity data infrastructures, knowledge graphs, ontology engineering, spatial Linked Data, and federated SPARQL querying. This body of work motivates the focus on high-quality observation data, RDF-based biodiversity modeling, GeoSPARQL representation, and query-time integration with external knowledge graphs.

2.1 Biodiversity Data Infrastructures and Knowledge Graphs

Biodiversity informatics has emphasized the need to mobilize and improve biodiversity observation data. Wetzel et al. [14] discuss gaps in European biodiversity observation data, while Güntsch et al. [6] identify essential functions of national biodiversity data infrastructures. In the Netherlands, ARISE supports species recognition and biodiversity monitoring [12], and van Strien et al. [10] address the improvement of citizen science plant data for long-term monitoring. OpenBiodiv is a closer linked data predecessor [9], which extracts taxonomic information from scholarly articles and taxonomic treatments, represents it in RDF using OpenBiodiv-O, links scientific names to the GBIF taxonomic backbone, and exposes the resulting biodiversity knowledge graph through SPARQL. However, OpenBiodiv focuses on literature-derived taxonomic knowledge such as treatments, taxonomic name usages, and publication metadata. The Nature FIRST Knowledge Graph is another biodiversity-related KG example [1], while Darwin Core provides the occurrence data vocabulary basis for WLW [15]. Darwin Core does not cover all NDFF-specific fields, such as conservation policy status, monitoring protocol, spatial grid identifiers, species group classification, and location blurring levels.

2.2 Ontology Engineering for NDFF Observation Data

General biodiversity and environmental resources such as Darwin Core, NCBI-Taxon, and ENVO support occurrence records, taxonomic classification, and environmental entities, but they do not directly model Dutch conservation policy status, monitoring protocols, spatial grid cells, species groups, or location blurring levels. This mirrors the situation in OpenBiodiv, which reuses existing vocabularies but introduces OpenBiodiv-O for taxonomic literature-specific concepts [9]. The WLW ontology was developed following SABIO [4], a methodology that structures ontology engineering into specification, conceptualization, formalization, implementation, and evaluation phases, grounding each design decision in competency questions derived from the target dataset.

2.3 Spatial Linked Data and GeoSPARQL

GeoSPARQL provides a standard vocabulary and query framework for representing and querying geospatial data on the Semantic Web, including geometry classes, spatial relations, and filter functions such as `geof:sfWithin` and `geof:distance` [3]. WLW uses GeoSPARQL to attach observation polygons as `geo:geoJSONLiteral` values and to perform spatial filtering against administrative boundaries retrieved from external endpoints.

Dutch governmental spatial Linked Data provides a relevant integration context. Folmer et al. [5] discuss the business rationale for Linked Data at the Netherlands' Kadaster Data Platform, showing how RDF, URIs, SPARQL, and Linked Data principles support Dutch governmental geospatial data integration. The Kadaster Knowledge Graph exposes authoritative Dutch geospatial data, including addresses, buildings, topography, cadastral parcels, and administrative areas.⁴

2.4 Federated SPARQL and Knowledge Graph Integration

Federated SPARQL enables query-time integration across distributed knowledge graphs without fully materializing external data locally. In SPARQL 1.1, the `SERVICE` clause delegates part of a query to a remote endpoint. In WLW, local scientific names are matched to Wikidata taxon names via `wdt:P225`, while broader taxonomic groups are retrieved by traversing the parent-taxon relation `wdt:P171*`.

Prior work highlights both the value and difficulty of such federation. Bolleman et al. [2] present SPARQL query pairs over bioinformatics knowledge graphs that include federated queries, and show that federated SPARQL requires users to understand multiple schemas and joins across endpoints. Thiery et al. [11] discuss distributed research data knowledge graphs and the challenges of federated querying across Wikidata, Wikibase instances, and other Linked Open Data resources. Hanski et al. [7] analyze Wikidata query federation and document

⁴ <https://data.kkg.kadaster.nl/>

practical trade-offs including high HTTP request counts, endpoint execution budgets, and timeout behaviors.

Overall, WLW is positioned at the intersection of biodiversity observation data, ontology engineering, spatial RDF, and federated SPARQL integration. Unlike biodiversity knowledge graphs focused mainly on literature-derived taxonomic knowledge, WLW represents regional Dutch NDFF occurrence records as a queryable RDF knowledge graph with GeoSPARQL geometries, NDFF-specific ontology modeling, and targeted federation with external taxonomic and cadastral knowledge.

3 Approach

3.1 Source Data and Vocabulary Mapping

The NDFF holds over 50 million occurrence records for the Netherlands, published at full coordinate precision. Observations for the Twente region were extracted by overlaying the Twente boundary with the NDFF grid. Data is requested and delivered per grid cell via email from NDFF; 81 such downloads were merged into a single dataset of 747,104 observation features covering 2025. A representative tile citation is given in [8].

Each feature contains 26 properties, of which the most relevant are listed in Table 1. Notable fields include *Identiteit*, a SHA-256 hash serving as a stable occurrence identifier, and *Hoknummer*, the enclosing 5×5 km grid cell identifier. *Beleidsstatus* encodes multiple conservation policy flags as a pipe-separated string, and *Vervaging* records the level of coordinate blurring applied for species protection.

Table 1. Key Properties in the NDFF GeoJSON Schema

Dutch Field	Translation	Type
Identiteit	Occurrence ID	SHA-256 hash
Wetenschappelijke naam	Scientific name	free text
Naam soort	Common name	free text
Soortgroep	Species group	enum
Periode start/stop	Observation period	datetime
Hoknummer	Grid cell ID	identifier
Bronhouder	Data provider	enum
Protocol	Monitoring protocol	enum
Beleidsstatus	Policy status	pipe-sep. enum
Vervaging	Blurring level	enum

Vocabulary reuse was prioritized to maximize interoperability with other Darwin Core-based datasets. Three vocabularies were selected. Darwin Core [15] provided the primary occurrence-level terms: *Wetenschappelijke naam*, *Naam soort*, *Aantal*, and *Identiteit* map to `dwc:scientificName`, `dwc:vernacularName`,

`dwc:individualCount`, and `dwc:occurrenceID` respectively. The *Periode start* and *Periode stop* fields were combined into `dwc:eventDate`, encoded as an ISO 8601 interval when the timestamps differ and as a single timestamp when identical. Year-level intervals of the form 2025-01-01T00:00:00/2025-12-31T23:59:59 occur when the observation date is deliberately coarsened to protect vulnerable species. Dublin Core Metadata Terms [13] was used for provenance, recording the NDFF as `dcterms:publisher` and the contributing platform, *waarneming.nl* or *Telmee.nl*, as `dcterms:source`. GeoSPARQL [3] was used to attach each observation’s polygon geometry as a `geo:geoJSONLiteral` via `geo:geometry`.

The triplification was implemented in Python using RDFLib. Each observation is a named individual of type `dwc:Occurrence` identified by a URI of the form `wlw:observation{id}`. The dataset is serialized as Turtle and published in the project repository. The full filtered dataset is loaded into an Apache Jena Fuseki 6.0 triplestore.

Eight source fields had no adequate mapping in the surveyed vocabularies: *Hoknummer*, *Beleidsstatus*, *Vervaging*, *Protocol*, *Soortgroep*, *Schaal*, *Telonderwerp*, and *Oorsprong*. These fields carry information central to the biodiversity use case, covering conservation policy status, monitoring protocol, grid cell location, and blurring level, and their absence from the occurrence triples motivates the ontology layer described next.

3.2 Domain Ontology Design

The ontology is designed as an extension layer over the triplified occurrence dataset. Both are loaded into the same Fuseki triplestore. The occurrence triples provide observation-level facts, while the ontology provides semantic structure for the eight fields that existing vocabularies could not represent. The integration mechanism is `dwc:scientificName`, which appears in both layers as a plain string literal and serves as the SPARQL join bridge, allowing the full observation dataset to be semantically enriched without modification to the occurrence triples.

The ontology was developed following the SABiO methodology [4]. The specification phase was driven by six competency questions, labelled CQ1 through CQ6, one per unmapped field. CQ1 and CQ3 together address *Beleidsstatus*, covering Red List status and EU Directive membership respectively. CQ2, CQ4, CQ5, and CQ6 address grid cell location, monitoring protocol, blurring level, and species group through *Hoknummer*, *Protocol*, *Vervaging*, and *Soortgroep* respectively.

The conceptual model, shown in Fig. 1, centers on `wlw:Observation`, connected through object properties to classes for species, observation area, period, monitoring protocol, habitat, and dataset. Two design decisions deserve explanation.

First, `PolicyStatus` is modeled as a parent class with three mutually disjoint subclasses: `RedListStatus`, `DirectiveStatus`, and `InvasiveStatus`. The source *beleidsstatus* field combines values from three distinct legal frameworks,

namely the Dutch Red List, the EU Birds and Habitats Directives, and the EU Invasive Species Regulation, in a single pipe-separated string. Subclassing allows queries to filter by status category using RDF type constraints rather than string comparison, and allows a reasoner to infer that any species carrying a `RedListStatus` individual is also an instance of `PolicyStatus`.

Second, `ObservationArea` is introduced as an intermediate class between `Observation` and `GridCell` to separate the observation event from its spatial footprint. `ObservationArea` carries the geometry, the grid cell reference via `wlw:intersectsGridCell`, the regional reference via `wlw:intersectsRegion`, and the blurring level via `wlw:hasBlurringLevel`.

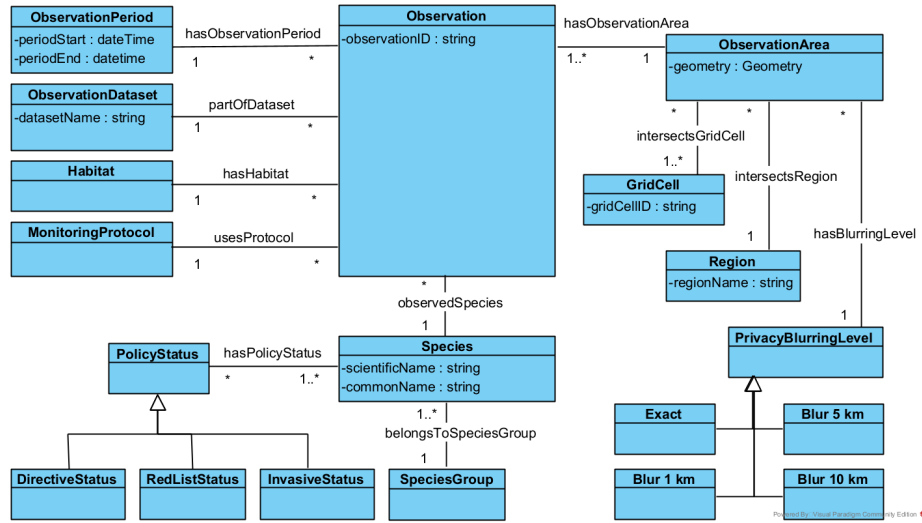


Fig. 1. UML class diagram of the Who Lives Where ontology. `PolicyStatus` has three mutually disjoint subclasses: `RedListStatus`, `DirectiveStatus`, and `InvasiveStatus`.

The ontology is implemented in OWL 2 using Turtle serialization under the `wlw:` namespace, defining 15 classes, 11 object properties, and 13 datatype properties. Controlled vocabulary individuals are defined directly in the ontology as fixed NDFS enumerations: 4 individuals for `PrivacyBlurringLevel`, 4 for `RedListStatus`, 3 for `DirectiveStatus`, and 2 for `InvasiveStatus`. The principal axioms are listed in Table 2. The ontology was validated with the Hermit reasoner in Protégé 5 with no inconsistencies detected.

Table 2. Formal Axioms in the WLW Ontology

ID	Axiom	OWL construct
A1	RedListStatus subClassOf PolicyStatus	rdfs:subClassOf
A2	DirectiveStatus subClassOf PolicyStatus	rdfs:subClassOf
A3	InvasiveStatus subClassOf PolicyStatus	rdfs:subClassOf
A4	RedListStatus, DirectiveStatus, InvasiveStatus mutually disjoint	owl:AllDisjointClasses
A5	All 12 core classes mutually disjoint	owl:AllDisjointClasses

Instances are typed as both `wlw:Observation` and `dwc:Occurrence`, ensuring compatibility with the occurrence layer. A Python triplification script generates ontology-structured triples for the full filtered dataset. Listing 1.1 aggregates observations by species group via the scientific name bridge; it returned 27 distinct groups in 0.17 seconds, confirming that the ontology enrichment propagates across the full corpus without modification to the occurrence triples.

```
PREFIX wlw: <https://2526-ldsw-g9.github.io/Who-Lives-Where/data.ttl#>
PREFIX dwc: <http://rs.tdwg.org/dwc/terms/>

SELECT ?speciesGroup (COUNT(?obs) AS ?count)
WHERE {
  ?obs a dwc:Occurrence ;
       dwc:scientificName ?scientificName .
  ?species a wlw:Species ;
           wlw:scientificName ?scientificName ;
           wlw:belongsToSpeciesGroup ?group .
  ?group wlw:groupName ?speciesGroup .
}
GROUP BY ?speciesGroup
ORDER BY DESC(?count)
```

Listing 1.1. Observation counts per species group, aggregated via the scientific name bridge

3.3 Federated Query Architecture

The triplestore is extended through query-time federation with two external endpoints: the Wikidata SPARQL endpoint for taxonomic knowledge, and the KKG SPARQL endpoint for Dutch administrative geometries. Both are accessed via SPARQL `SERVICE` clauses at query time rather than through materialized `owl:sameAs` links, avoiding the maintenance overhead of keeping alignments current with evolving external graphs.

Wikidata is linked through scientific name. Local `dwc:scientificName` values are matched to Wikidata taxon names via `wdt:P225`. The parent-taxon property `wdt:P171*`, traversed with zero-or-more hops, enables taxonomy-aware queries that retrieve all descendants of a given taxon regardless of depth, for ex-

ample all species within the order Strigiformes, without importing any part of Wikidata locally.

KKG is linked through geometry. Administrative region boundaries are retrieved from the KKG endpoint and combined with local observation geometries using `geof:sfWithin`. Figure 2 shows a subset of observations visualized in the KKG Geo viewer. The full dataset cannot be rendered in a single federated query because the KKG endpoint enforces a query timeout, requiring a `LIMIT` that captures only part of the distribution.

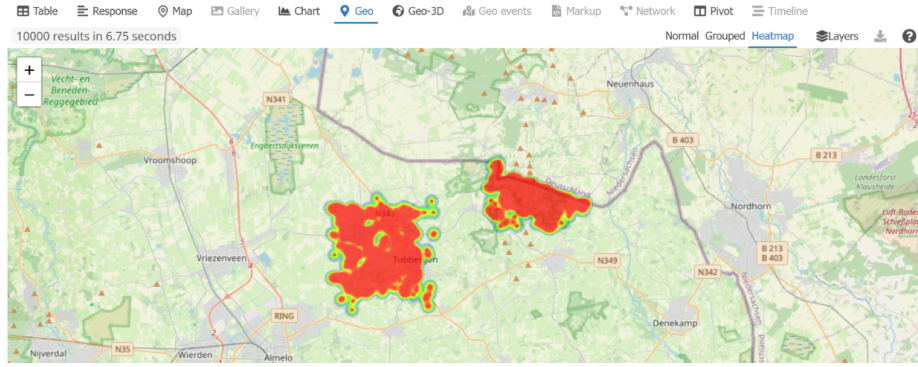


Fig. 2. Spatial density of biodiversity observations in the Twente region, visualized in the KKG Geo viewer. Only a subset of observations is shown due to the KKG endpoint’s query timeout.

Listing 1.2 shows a federated query combining all three sources: it retrieves the Tubbergen municipality boundary from KKG, resolves owl taxa through Wikidata’s `wdt:P171*` hierarchy from the order Strigiformes, and spatially filters local observations using `geof:sfWithin`.

4 Evaluation

The evaluation mainly considers three aspects of WLW: competency question coverage, structural validation, and analytical query utility. Competency question coverage focuses on whether the ontology layer covers the domain specific NDDF fields that could not be represented with existing vocabularies. Structural validation checks whether the integrated graph conforms to the expected ontology structure. Analytical query utility demonstrates whether the graph supports practical analysis through SPARQL and GeoSPARQL queries.

```

PREFIX geof: <http://www.opengis.net/def/function/geosparql/>
PREFIX imxgeo: <http://modellen.geostandaarden.nl/def/imx-geo#>

SELECT ?commonNameEn (COUNT(?uri) AS ?count)
WHERE {
  SERVICE <https://data.kkg.kadaster.nl/service/sparql> {
    ?municipality a imxgeo:Gemeentegebied ;
    imxgeo:naam "Tubbergen" ;
    geo:hasGeometry/geo:asWKT ?muniGeom .
  }

  SERVICE <https://query.wikidata.org/sparql> {
    SELECT ?taxon ?name ?commonNameEn WHERE {
      ?taxon wdt:P171* wd:Q25222 ;
      wdt:P225 ?name .
    }
    OPTIONAL {
      ?taxon rdfs:label ?commonNameEn .
      FILTER(LANG(?commonNameEn) = "en")
    }
  }
}

?uri a dwc:Occurrence ;
dwc:vernacularName ?commonName ;
wlv:hasObservationArea/geo:hasGeometry/geo:asWKT ?geometry ;
dwc:scientificName ?name .

FILTER(geof:sfWithin(?geometry, ?muniGeom))
}
GROUP BY ?commonNameEn
ORDER BY DESC(?count)

```

Listing 1.2. Federated query: owl species most frequently observed in Tubbergen, combining local observations, Wikidata taxonomy, and KKG geometry

4.1 Competency Question Coverage

The six competency questions (CQ1-CQ6) defined during ontology engineering serve as the acceptance criteria for the ontology layer. They cover the NDFF specific information that could not be adequately represented with Darwin Core, Dublin Core Metadata Terms, or GeoSPARQL in the occurrence layer, including conservation policy status, grid cell, habitats directive status, monitoring protocol, location blurring level, and species group.

Coverage was evaluated through SPARQL queries over the integrated graph. The ontology layer makes the selected domain-specific information jointly queryable via the scientific name bridge. Listing 1.3 confirms that this enrichment propagates to the full corpus. Executed against the triplestore, it returned 27 distinct species groups in 0.17 seconds without modifying the original occurrence triples.

4.2 SHACL Validation

SHACL validation was used to evaluate whether the integrated graph conforms to the ontology structural expectations. Shapes were defined for the main classes connecting the occurrence and ontology layers, covering occurrences, species,

```

PREFIX wlw: <https://2526-ldsw-g9.github.io/Who-Lives-Where/data.ttl#>
PREFIX dwc: <http://rs.tdwg.org/dwc/terms/>
SELECT ?speciesGroup (COUNT(?obs) AS ?count)
WHERE {
  ?obs a dwc:Occurrence ;
       dwc:scientificName ?scientificName .
  ?species a wlw:Species ;
           wlw:scientificName ?scientificName ;
           wlw:belongsToSpeciesGroup ?group .
  ?group wlw:groupName ?speciesGroup .
}
GROUP BY ?speciesGroup
ORDER BY DESC(?count)

```

Listing 1.3. Query 2: Observation counts per ontology layer species group

observation areas, grid cells, regions, monitoring protocols, and policy status resources.

The shapes encode core structural expectations of the integrated graph. Listing 1.4 shows the representative SHACL shape for species validation.

```

wlw:SpeciesShape
  a sh:NodeShape ;
  sh:targetClass wlw:Species ;

  sh:property [
    sh:path wlw:scientificName ;
    sh:minCount 1 ;
    sh:maxCount 1 ;
    sh:datatype xsd:string ;
    sh:message "Each species must have exactly one scientificName." ;
  ] ;

  sh:property [
    sh:path wlw:hasPolicyStatus ;
    sh:minCount 0 ;
    sh:nodeKind sh:IRI ;
    sh:or (
      [ sh:class wlw:PolicyStatus ]
      [ sh:class wlw:RedListStatus ]
      [ sh:class wlw:DirectiveStatus ]
      [ sh:class wlw:InvasiveStatus ]
    ) ;
    sh:message "If present, hasPolicyStatus must point to a valid policy status resource." ;
  ] .

```

Listing 1.4. Representative SHACL shape for species validation

Shapes were complemented by targeted SPARQL sanity checks covering occurrence-to-species links, species scientific names, observation area grid cell references, and policy status targets. A representative shape was also tested in the Zazuko SHACL validator to confirm standard conformance report generation. No violations were found across all examined constraints.

4.3 Analytical Query Results

The analytical queries evaluate whether the integrated graph supports practical biodiversity exploration beyond direct triple lookup. The selected questions cover location precision, seasonal observation patterns, federated taxonomic retrieval, and spatial proximity. Regarding location accuracy, 84.2% of observations carry exact coordinates while the remaining 15.8% are spatially blurred for species protection. Figure 3 shows the monthly distribution of observations. January stands out with approximately 160,000 records, roughly twice the April and May counts, attributable to structured counting surveys at the start of the year rather than a genuine ecological signal. Figure 4 shows owl species observed in Enschede via the federated query of Listing 1.2. The Tawny Owl dominates with approximately 140 records while the Western Barn Owl accounts for fewer than ten, suggesting the Tawny Owl is well established in the urban landscape whereas the Barn Owl remains comparatively rare.

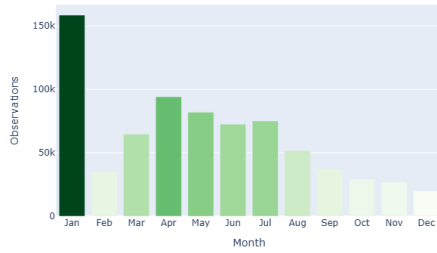


Fig. 3. Monthly distribution of observations in Twente. The January peak reflects structured annual surveys.

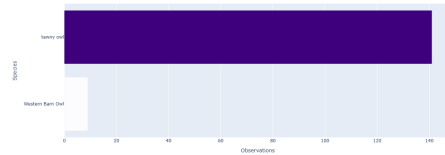


Fig. 4. Owl species in Enschede (AQ3), via the federated query in Listing 1.2.

Regarding the proximity of selected species to the queried address (AQ4), Figure 5 shows all three assessed species were observed within relatively short distances. The common pipistrelle had the nearest recorded observation at 0.09 km, followed by the Asian hornet at 0.16 km and Japanese knotweed at 0.52 km. The close proximity of the Asian hornet is particularly noteworthy given its status as an invasive species with documented ecological impacts in the Netherlands.

5 Discussion and Conclusion

Augmenting a flat occurrence graph with a separate ontology layer proved effective as a general integration pattern. Semantic enrichment was added without re-triplication, and the scientific name bridge allowed ontology classifications to propagate across the full observation corpus. This pattern is applicable to any structured biodiversity dataset with partial Darwin Core coverage, where the

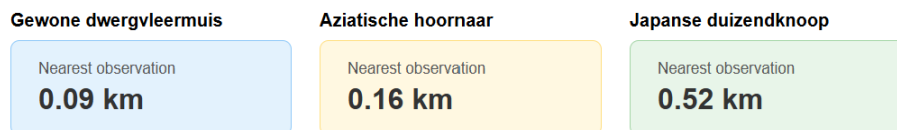


Fig. 5. Minimum distance from a queried building address to the nearest recorded observation of three target species (AQ4). Building geometry is retrieved from KKG; `geof:distance` is evaluated over local observation geometries.

occurrence layer can be published once and enriched incrementally as domain knowledge is formalized.

Modeling `PolicyStatus` as a class hierarchy rather than a flat enumeration was the most consequential design decision. Queries over Red List, Directive, and Invasive status can be formulated using RDF type constraints rather than string matching, and a reasoner can infer superclass membership without additional triples. The gain in expressiveness directly reflects the structure of the source data, where a single pipe-separated field combines values from three legally distinct frameworks.

Query-time federation via SPARQL `SERVICE` proved a practical strategy for linking to both Wikidata and KKG. Retrieving taxonomic hierarchies dynamically through `wdt:P171*` avoids the maintenance overhead of materialized `owl:sameAs` links, which would need updating whenever Wikidata taxonomy is revised. The trade-off is that federated queries inherit the availability and timeout constraints of the external endpoints, as illustrated by the `LIMIT` required for KKG visualization. SHACL validation confirmed structural consistency of the integrated graph, though it checks graph structure rather than semantic correctness. Domain-level invariants such as correct policy flag assignment require additional manual verification.

Several limitations should be noted. Scientific names are not globally stable identifiers, and taxonomy revisions can silently break the join bridge between layers. The dataset covers only observations from 2025 in Twente, leaving temporal and geographic generalizability untested. The NDDFF download workflow is manual and requires institutional access credentials, which limits reproducibility. Federated queries depend on the continued availability of the Wikidata and KKG endpoints.

Taken together, the results show that a triplification pipeline built on Darwin Core and GeoSPARQL, extended with a purpose-built domain ontology and federated to external knowledge graphs, can support spatial, taxonomic, and policy-driven queries over large-scale biodiversity data. The six competency questions are fully answerable against the integrated graph, the aggregation query returns 27 species groups in 0.17 seconds, and SHACL validation reports no structural violations. Future work should replace the scientific name bridge with stable taxon identifiers such as Wikidata entity URIs, formalize the GeoSPARQL 1.1

alignment by making `wlw:ObservationArea` a subclass of `geo:Feature`, extend SHACL coverage to the full triplified dataset, and explore integration with additional KKG layers such as BAG buildings and BGT land use for planning-oriented queries. Connecting the pipeline to permit and land-use workflows would make conservation constraints directly queryable within enterprise planning systems, which is the use case that originally motivated this work.

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